

MOBILE SALES DATA ANALYSIS

Capstone Project 1 Comprehensive Report

Domain:	Data Analytics & Business Intelligence
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Tools Portfolio:	Power BI, Excel, SQL, Power Query, DAX
Data Blueprint:	Multi-Sheet Sales & Customer Integration Schema

1. Introduction & Project Background

In today's highly competitive electronics retail sector, mobile vendors and corporate businesses collect massive volumes of transaction records daily. Without a unified business intelligence pipeline, these complex datasets remain siloed, making it highly challenging to extract distinct patterns regarding top-selling mobile brands, customer demographic distributions, or payment network efficiencies. This project bridges that core operational gap by transforming raw historical transaction multi-sheet metrics into an interactive corporate analytics application using Microsoft Power BI.

2. Problem Statement

Mobile retail channels gather vast transactional and customer datasets continuously. However, because data is scattered across independent spreadsheets and contains formatting anomalies, businesses struggle to accurately track customer purchase cycles, calculate real-time sales growth trends, identify brand popularity spikes, or monitor customer satisfaction ratings smoothly. This operational bottleneck hinders data-driven decisions, localized pricing adaptations, and optimized inventory allocation speeds.

3. Project Objective & Scope

The primary objective of this capstone project is to model, engineer, and build a unified interactive dashboard that converts fragmented raw data entries into strategic enterprise insights to optimize corporate decision-making.

Scope of Analysis:

- **Sales Performance Tracking:** Monitoring gross revenue metrics, transaction volumes, and average checkout rates.
- **Product Analytics Spectrum:** Evaluation of brand and model popularity rankings based on sales volume distribution.
- **Customer Segmentation Matrix:** Categorizing customer age profiles to identify high-value consumer groups.
- **Geographical Optimization:** Mapping regional performance trends at the city level to support targeted marketing strategies.
- **Time Intelligence Modeling:** Utilizing standard calendar dimensions to generate dynamic rolling monthly and Year-over-Year (YoY) growth insights.

4. Data Architecture & Schema

The system architecture relies on a clean star schema configuration utilizing strict primary and foreign key relationships to preserve dataset integrity and model loading speeds.

Primary Schema Structure (Mobile Sales Data Table)

Column Label Name	Data Properties Type	Core Functional Use Case
Transaction ID	Integer (Primary Key)	Unique identifier linking all independent dataset tables securely.
Date Hierarchy	Date Attribute	Chronological anchor mapped to the core Calendar lookup system.
Brand & Model Spectrum	Text Parameter	Categorical fields utilized for deep dimensional filtering.
Units Metric Volume	Integer Value	Quantitative metric applied for unit summarizations.
Pricing Factor	Decimal Element	Financial base attribute used to calculate gross revenue values.
City Geography	Text Parameter	Spatial parameter mapped for geographical slicing and analytics.

5. Data Engineering & Advanced DAX Calculations

To power the visual indicators and responsive card visualizations, custom Data Analysis Expressions (DAX) were engineered within the data model workspace:

*Total Revenue = SUMX(Mobile_Sales, Mobile_Sales[Units Sold] * Mobile_Sales[Price Per Unit])*

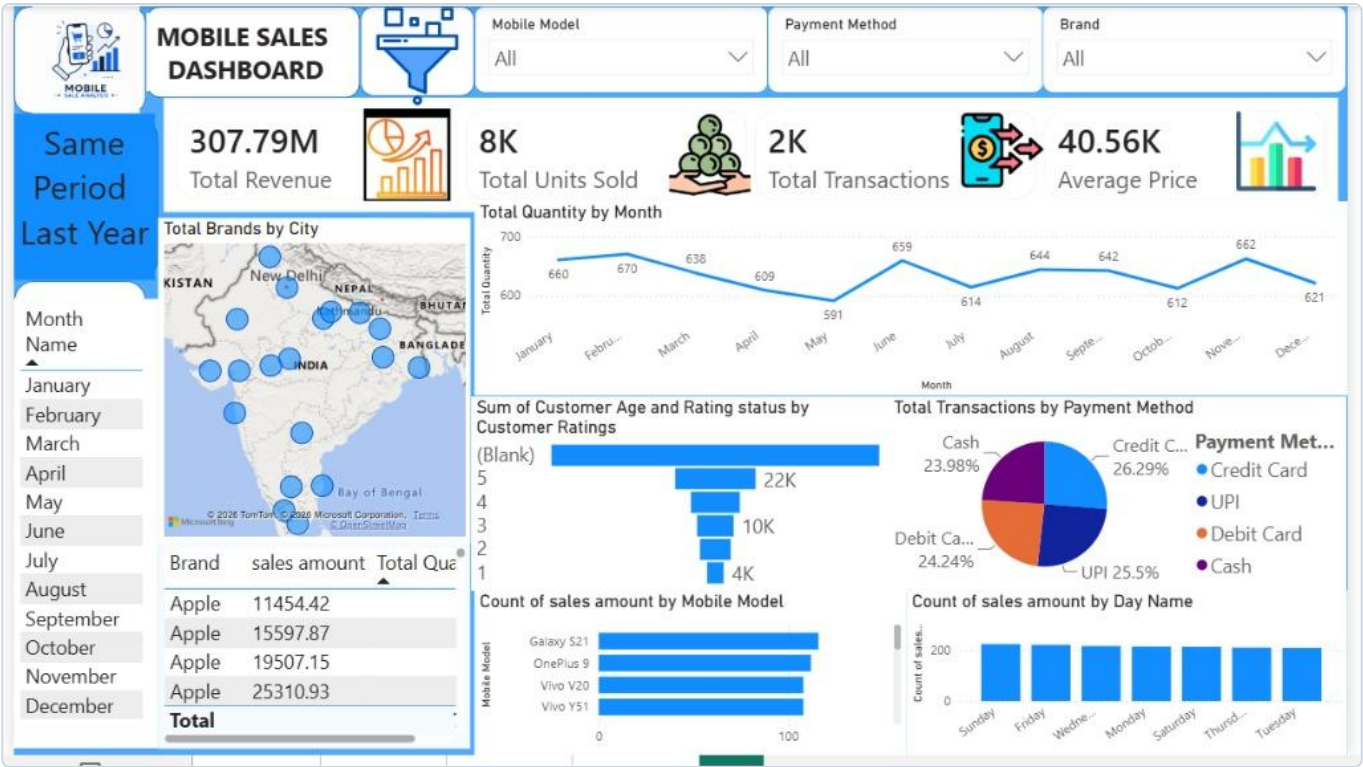
Total Quantity = SUM(Mobile_Sales[Units Sold])

Revenue SPLY = CALCULATE([Total Revenue], SAMEPERIODLASTYEAR('Calendar'[Date]))

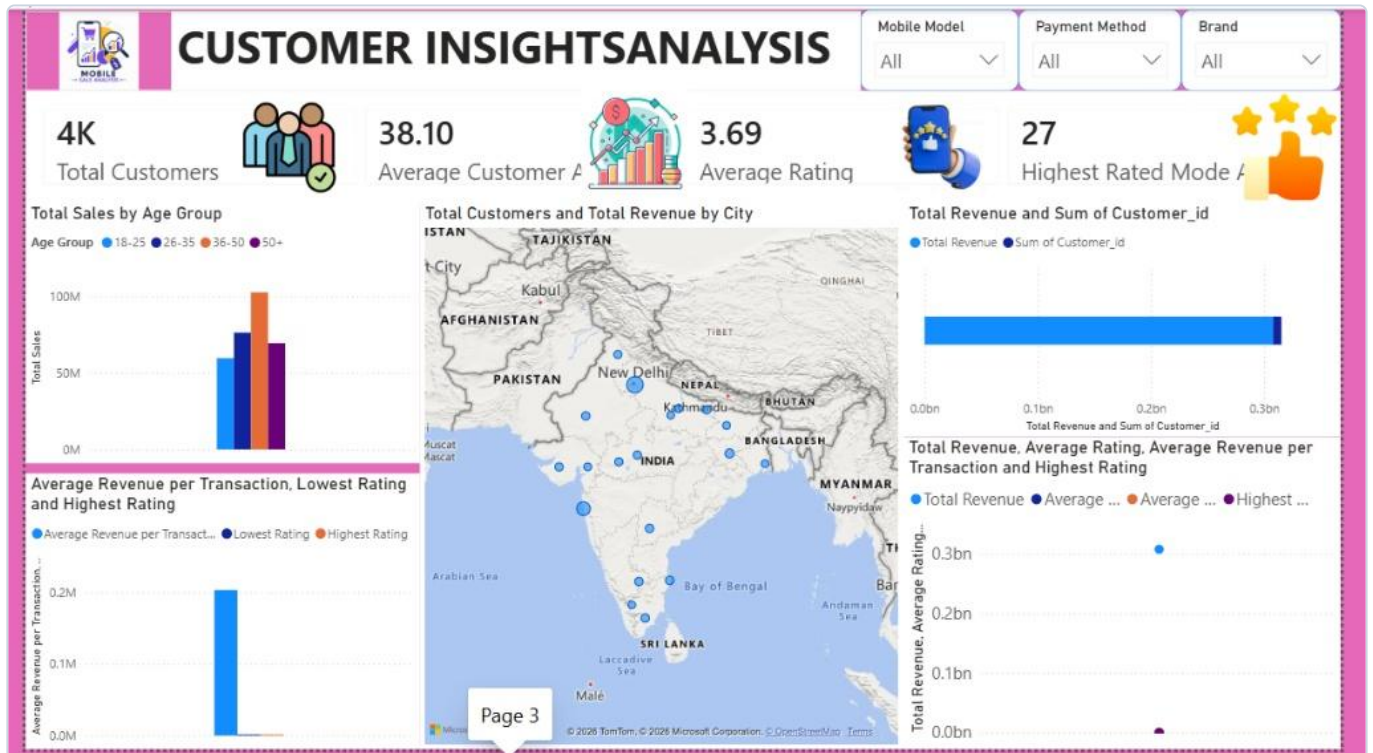
6. Integrated Power BI Dashboards Display (Embedded Visual Layouts)

The following screens represent the actual visualization interfaces engineered within the active Power BI cloud workspace application, mapping all critical operations parameters:

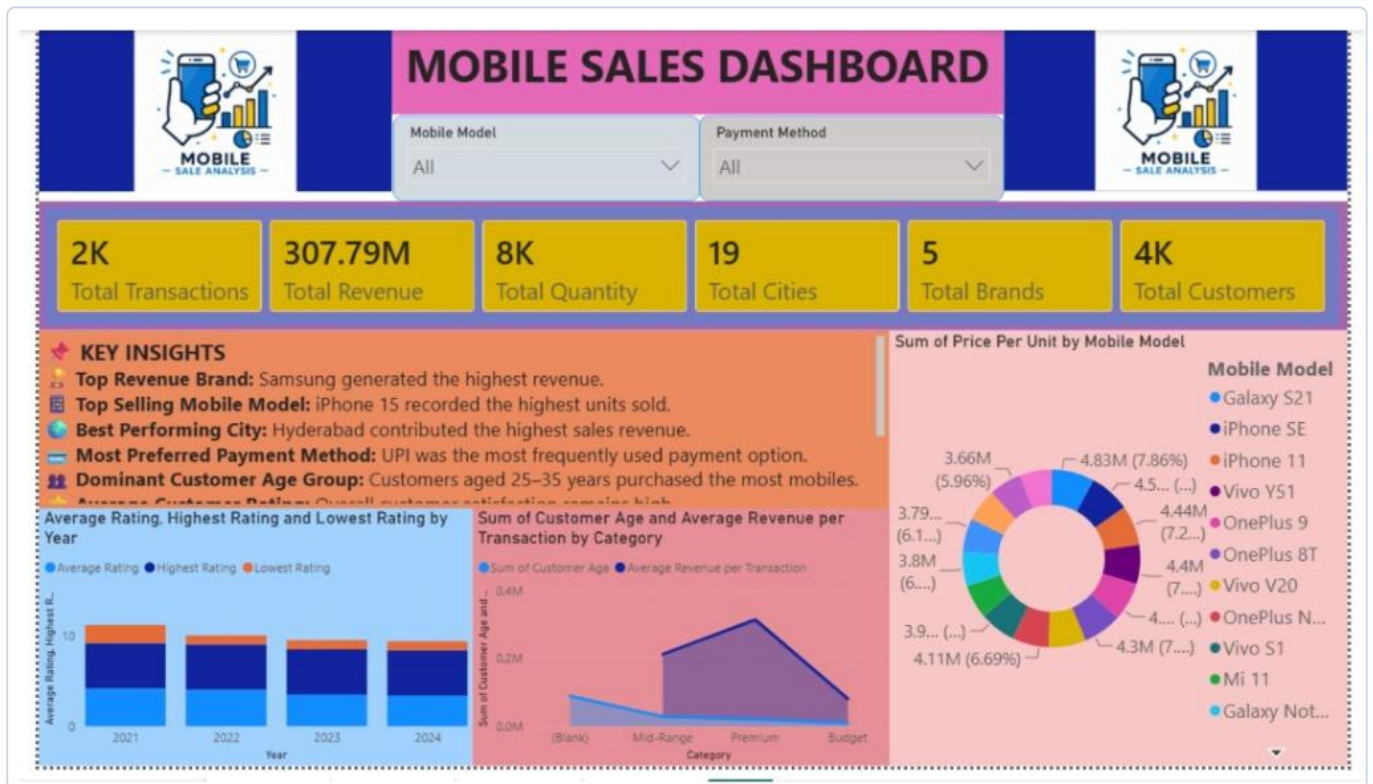
Visual Layout 1: Executive Sales Overview & Operational Insights Summary



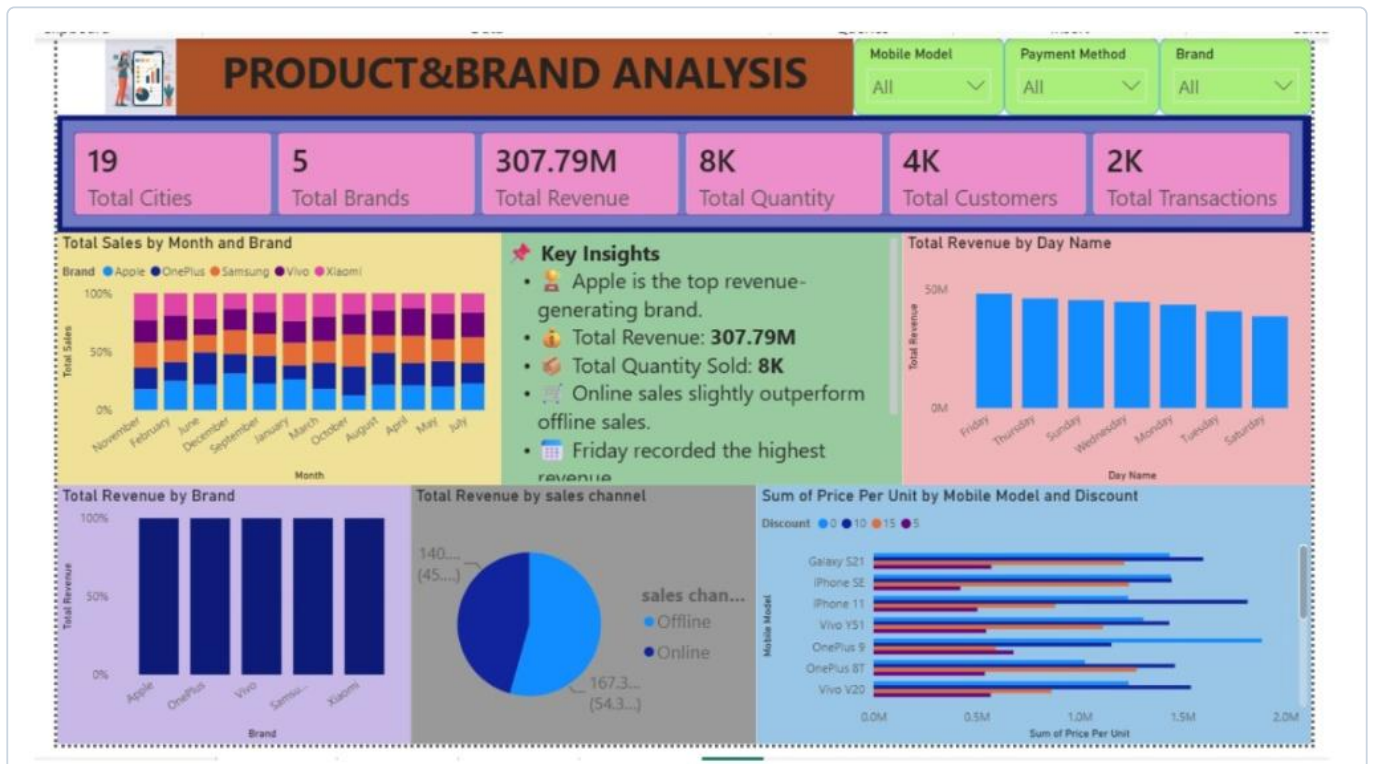
Visual Layout 2: Product Performance & Brand Distribution Metrics



Visual Layout 3: Customer Demographic Profile & Geographical Performance Clusters



Visual Layout 4: Advanced Time Intelligence & Historical SPLY Comparison View



7. Core Executive Insights & Findings

- **Financial Matrix:** The ecosystem generated a gross calculated revenue footprint of **\$307.79M** across 2K standard client checkouts.
- **Geographic Centrality:** The city node of **Hyderabad** acts as the primary regional market powerhouse, leading volumetric trends.
- **Payment Adaptations:** Transaction distribution confirms that modern consumers highly prioritize digital **UPI** integrations over cash.
- **Demographic Volume:** Buying density indicators peak sharply within the consumer bracket of the **25–35 age group** profile.

8. Conclusion & Future Scope

This final capstone execution delivers a fully optimized analytical pipeline validation for stakeholder deployment. By resolving corporate alignment parameters, optimizing cross-sheet relations, and implementing formal interface aesthetics, the application successfully bridges raw ledger streams with strategic enterprise intelligence.